Stem society

CS : DIGICON

The Message Summit ; Rebirth Of the Phoenix

INRTODUCTION

STEM DIGICON CHALLENGES STUDENTS FROM GRADES 8-11 TO APPLY THEIR DIGITAL LITERACY, LOGICAL THINKING AND PROBLEM-SOLVING SKILLS. PARTICIPANTS WILL COMPETE IN BOTH TEAM-BASED AND INDIVIDUAL CHALLENGES, TESTING THEIR KNOWLEDGE OF COMPUTER SCIENCE, CYBERSECURITY, NUMBER SYSTEMS, AI-GENERATED CONTENT AND GRAPHIC DESIGNING. THROUGH INTERACTIVE ROUNDS AND A WRITTEN TEST, STUDENTS WILL PUT THEIR TECHNICAL KNOWLEDGE AND PRACTICAL SKILLS TO THE TEST. THIS STUDY GUIDE OUTLINES THE KEY CONCEPTS AND SKILLS REQUIRED TO SUCCEED IN ALL ROUNDS

STUDY MATERIAL

1. STUDENTS REGISTERING AS STEM CARTEL NEED TO BE ABLE TO:

1. COMPUTER HARDWARE AND ARCHITECTURE
   - CPU, RAM, ROM AND STORAGE
   - INPUT AND OUTPUT DEVICES
   - HOW A COMPUTER PROCESSES INFORMATION

2. NUMBER SYSTEMS AND DATA REPRESENTATION
   - BINARY, DECIMAL AND HEXADECIMAL
   - BINARY CONVERSIONS
   - BITS, BYTES AND DATA UNITS
   - ASCII AND UNICODE
   - IMAGE AND SOUND REPRESENTATION

3. CYBERSECURITY AND DIGITAL SAFETY
   - MALWARE AND VIRUSES
   - FIREWALLS
   - PHISHING AND ONLINE PRIVACY

4. CONTROL SEQUENCESSTUDENTS WILL HAVE TO MEMORIZE A LIST OF WINDOW KEY SHORTCUTS.

CATEGORY OVERVIEW

DIGICON :

THE CATEGORY CONSISTS OF TWO MAIN ROUNDS: SQUAD BATTLE (TEAM ROUND) AND SOLO SHOWDOWN (INDIVIDUAL ROUND). PARTICIPANTS WILL FIRST COMPETE IN TEAMS THROUGH FOUR INTERACTIVE STATIONS, WITH THE TOP-PERFORMING TEAMS ADVANCING TO THE SECOND ROUND. THE FINAL ROUND WILL BE AN INDIVIDUAL WRITTEN COMPETITION, WHERE PARTICIPANTS WILL EARN CODE POINTS BASED ON THEIR PERFORMANCE. THE COMPETITION CARRIES A TOTAL OF 100 CODE POINTS, AND THE TWO PARTICIPANTS WITH THE HIGHEST OVERALL SCORES WILL BE DECLARED THE WINNERS

ENIGMA :

.THE CATEGORY CONSISTS OF TWO MAIN ROUNDS: INVESTIGATING CASE FILES (TEAM ROUND) AND SCAVENGER HUNT STYLE QUIZ (INDIVIDUAL ROUND). PARTICIPANTS WILL INITIALLY REGISTER INDIVIDUALLY FOR MATHEMATICS OR INTEGRATED SCIENCES AND WILL THEN BE GROUPED INTO MIXED-SUBJECT TEAMS TO SOLVE A CASE FILE USING MATHEMATICAL AND SCIENTIFIC KNOWLEDGE. THE TOP-PERFORMING TEAMS WILL ADVANCE TO THE FINAL ROUND, WHERE EACH QUALIFYING PARTICIPANT WILL COMPETE INDIVIDUALLY IN A SUBJECT-SPECIFIC SCAVENGER HUNT BY SOLVING QUESTIONS WITHIN A GIVEN TIME LIMIT. MATHEMATICS AND INTEGRATED SCIENCES WILL HAVE SEPARATE COMPETITIONS, WITH A WINNER AND RUNNER-UP DECLARED FOR EACH CATEGORY.